
Patient-Aware Contrastive Learning Preserves Per-Patient Structure in RR-Interval Representations

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Abstract

Contrastive representation learning struggles on physiological signals when each subject contributes a distinct baseline pattern. If class differences overlap with subject differences, class-level objectives such as supervised contrastive learning tend to merge per-subject structure into a single per-class cluster, removing the individual variation that a model needs to generalize to unseen patients. We study this problem in the setting of Paroxysmal Atrial Fibrillation (PAF) detection from RR-interval (RRI) sequences and propose a *patient-aware contrastive objective* that forms positive pairs only from same-patient, same-class segments, preserving each patient’s own sinus rhythm (SR) baseline while still pushing the two classes apart. Examining the learned embeddings directly, our objective achieves the most consistent per-patient SR structure (cohesion 0.850 vs. 0.800 for supervised contrastive loss (SupCon) and 0.772 for binary cross-entropy (BCE)). We also identify that BCE produces the cleanest global class separation yet the most disordered per-patient structure. This is precisely why a linear probe trained on its features breaks down on unseen patients. On the IRIDIA-AF dataset, the resulting representation reaches a patient-independent Area Under the Receiver Operating Characteristic Curve (AUROC) of 0.989 ± 0.003 with $2.6\times$ lower seed variance than supervised contrastive baselines. These results highlight that per-subject geometric consistency, rather than global class separability, is key to robust cross-patient generalization.

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⁰Code - github.com/EML-Labs/pacl-rr-interval

1. Introduction

Contrastive representation learning is now the default for unlabeled and weakly-labeled time-series and physiological signals (Chen et al., 2020; Le-Khac et al., 2020; Wang & Isola, 2020; Khosla et al., 2021). However, its standard formulations assume that every same-class example is equally suitable as a positive. This assumption breaks down on data where each example carries both a class label y and a subject identifier p . When class differences are partly aligned with subject differences, instance-level objectives blend class and identity together. While class-level objectives such as supervised contrastive learning fold each subject’s distinct physiology into a single shared class cluster. The resulting embeddings look well-separated overall yet are inconsistent within each subject. Therefore, a linear probe trained on one cohort fails to transfer to another. Our position is that what generalizes across subjects is not how cleanly classes are separated overall, but how consistently each subject is represented.

We study how to build this property directly into the contrastive objective. We instantiate this question on RRI sequences for Atrial Fibrillation (AF) detection: every patient has a distinct SR baseline, AF dynamics vary across individuals, and generalizing to unseen patients is the main bottleneck. Andersen et al. reported sensitivity dropping from 98.98% to 86.04% on held-out patients (Andersen et al., 2019; De With et al., 2020; Joglar et al., 2024). Self-supervised and contrastive methods have shown promise for Electrocardiogram (ECG) signals (Grill et al., 2020; Liu et al., 2023; Hu et al., 2025; Sun et al., 2025; Chen et al., 2025), and patient-level objectives have been studied for unsupervised ECG pre-training (Kiyasseh et al., 2021; Diamant et al., 2022). What is missing is a positive-pair construction that is simultaneously class-aware and subject-aware, so it directly targets the per-class, per-subject structure that governs cross-patient transfer.

Contributions.

- A **patient-aware contrastive objective** that, for each anchor, treats only same-patient, same-class segments as positives, preserving each subject’s individual structure while still pulling apart classes.

- An **embedding geometry analysis** that explains the mechanism: our objective attains the highest per-patient SR cohesion among the compared losses. We also identify that BCE gives the cleanest global class separation yet the most disordered per-patient structure. This is consistent with its weaker transfer to new patients.
- **Downstream validation on AF detection** via frozen-encoder linear probing on IRIDIA-AF, reaching AU-ROC 0.989 ± 0.003 with $2.6 \times$ lower seed variance than supervised contrastive baselines.

Related work. Contrastive representation learning has progressed from instance discrimination objectives (Chen et al., 2020; Grill et al., 2020) to label-aware extensions (Khosla et al., 2021), with Wang & Isola (2020) describing the geometry of the learned representations through alignment and uniformity on the hypersphere. In the cardiac domain, several patient-level contrastive variants exist: **CLOCS** (Kiyasseh et al., 2021) contrasts ECG segments across space, time, and patients but is unsupervised and treats any same-patient pair as a positive; **PCLR** (Diamant et al., 2022) uses same-patient recordings as positives in a SimCLR-style instance task; **PMQ** (Sun et al., 2025) adds a patient memory queue to enrich intra-patient comparisons during pre-training. All three are *unsupervised* pre-training methods that use patient identity to define positives without using class labels. Our objective, in contrast, is a *supervised* one whose positive set is the intersection of class label and patient identity—it relies on class supervision to prevent supervised contrastive learning from collapsing each subject into a shared class cluster, while using subject identity to prevent unsupervised methods from blending class and identity. To our knowledge, no prior work applies this combined class-and-subject construction to PAF detection or analyses the resulting embedding geometry at the per-subject level.

2. Methodology

2.1. Dataset and Preprocessing

We use IRIDIA-AF (Gilon et al., 2023), comprising long-term single-lead ECG recordings from 167 patients with paroxysmal AF. Episodes are retained when AF duration ≥ 1 hr and the immediately preceding SR duration ≥ 4 hr. Splits are patient-level (119/24/24 train/val/test); after quality filtering, 154/27/24 episodes remain. RRI sequences are RobustScaler normalized per-patient using the first SR hour as the fit window, with classification windows drawn from a strictly disjoint subsequent hour (Appendix A and Figure 3 therein). The normalized stream is segmented with a sliding window of $W = 200$ beats and stride $S = 50$ beats. Physiologically implausible beats (< 200 ms or > 2000 ms) are discarded.

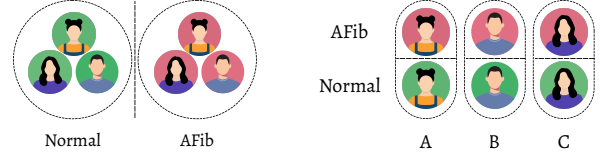


Figure 1. Standard supervised contrastive learning (left) treats all same-class segments as positives regardless of subject. The proposed patient-aware formulation (right) restricts positives to same-patient, same-class segments.

2.2. Patient-Aware Mini-Batch Sampling

Each mini-batch samples P patients, each contributing n SR and n AF windows ($B = 2nP$). A patient is eligible only if both its SR and AF pools contain at least n windows, guaranteeing intra-patient, intra-class positive pairs in every batch and enforcing class balance across patients.

2.3. A Patient-Aware Contrastive Objective

We formulate the objective as a generic template for subject-structured data, where each sample carries a class label y and a subject identifier p . The positive set is defined to be *intra-class and intra-subject*. SR/AF (PAF detection) is the instantiation used here. The construction applies wherever samples are grouped by subject and the within-subject variation is informative. For an anchor i with class y_i and subject p_i , the positive $\mathcal{P}(i)$ and negative $\mathcal{N}(i)$ sets are,

$$\mathcal{P}(i) = \{j \neq i \mid p_j = p_i \wedge y_j = y_i\}, \quad (1)$$

$$\mathcal{N}(i) = \{j \neq i \mid y_j \neq y_i\}. \quad (2)$$

Critically, same-class segments from different subjects are excluded from $\mathcal{P}(i)$, preventing the encoder from collapsing distinct individual SR baselines into a single shared prototype. This is the key departure from supervised contrastive learning (Khosla et al., 2021) (Figure 1). The per-anchor loss is the standard InfoNCE formulation,

$$\mathcal{L}_i = -\log \frac{\sum_{j \in \mathcal{P}(i)} \exp(s_{ij})}{\sum_{j \in \mathcal{P}(i)} \exp(s_{ij}) + \sum_{k \in \mathcal{N}(i)} \exp(s_{ik})}, \quad (3)$$

with $s_{ij} = \hat{z}_i^\top \hat{z}_j / \tau$ and learnable temperature τ ; the batch loss is $\mathcal{L} = \frac{1}{B} \sum_i \mathcal{L}_i$.

2.4. Encoder and Training

We use a lightweight multi-branch CNN backbone to demonstrate that the gains come from the loss, not from model capacity. This is to remain compatible with edge wearable hardware. The encoder consists of three parallel 1D-CNN

branches ($k \in \{3, 5, 7\}$) with stride convolutions (channels 16, 32, 64) (Szegedy et al., 2015), Group Normalization (Wu & He, 2018), and ReLU. Branch outputs are fused by a 1×1 convolution and pooled by a softmax temporal attention module (Shashikumar et al., 2018). A two-layer MLP projects to $\mathbb{R}^{128}$, and both projection and pooled outputs are ℓ_2 -normalized to the unit hypersphere (Wang & Isola, 2020). The contrastive loss acts on the projected embedding $\hat{\mathbf{e}}$. The linear probe is trained on the pooled output $\hat{\mathbf{z}}$. All three loss functions (Proposed, SupCon, BCE) are trained with the identical patient-aware sampler described in Section 2.3. The sampler is not varied across conditions. Training uses AdamW with a cosine scheduler. We use Optuna-TPE hyperparameter optimization (Akiba et al., 2019) to select the hyperparameters on the validation split (full configuration in Appendix B).

3. Experiments

Protocol. We adopt frozen encoder linear probing, a standard representation quality readout in contrastive learning (Wang & Isola, 2020; Chen et al., 2020; Khosla et al., 2021). Probe performance reflects the embedding geometry rather than classifier capacity. All experiments are repeated over five independent seeds and reported as mean $\pm$ std on the held-out patient-independent test split.

3.1. Embedding Geometry

We probe the embedding geometry directly. We compare the proposed objective against SupCon (Khosla et al., 2021) and BCE baselines, fixing the encoder, sampler, and probe. Only the temperature (when applicable) τ and learning rate is varied to find the best performance. We measure,

- Per-patient *class cohesion*, the mean cosine similarity of same-patient same-class embeddings.
- Global class separability via centroid distance, centroid cosine similarity, and compactness ratio
- Per-patient compactness ratio.

Full definitions are in Appendix C. Results are summarized in Table 1. SR cohesion is the primary metric because sinus rhythm is each patient’s individual baseline which is the reference point a cross-patient linear probe must extrapolate from when it encounters a new subject. High AF cohesion without corresponding transfer improvement confirms that SR placement consistency, not AF compactness, is the bottleneck.

Per-patient SR cohesion confirms the design. The proposed objective achieves the highest SR cohesion (0.850 vs. 0.800 SupCon vs. 0.772 BCE), evidence that intra subject positives preserve each subject’s own SR baseline rather

Table 1. Embedding-space metrics on the test set (mean $\pm$ std, 5 seeds). Per-patient SR cohesion is the primary metric. It directly measures the per-subject geometric consistency the objective targets. The bottom row reports downstream linear-probe AUROC. Bold = best per row. $\uparrow$ higher is better, $\downarrow$ lower is better.

Metric	Proposed	SupCon	BCE
SR cohesion $\uparrow$	0.850 $\pm$ 0.044	0.800 $\pm$ 0.029	0.772 $\pm$ 0.049
AF cohesion $\uparrow$	0.846 $\pm$ 0.048	0.921 $\pm$ 0.035	0.955 $\pm$ 0.019
Centroid sim. $\downarrow$	+0.238 $\pm$ 0.200	+0.155 $\pm$ 0.123	−0.717 $\pm$ 0.029
Centroid dist. $\uparrow$	1.034 $\pm$ 0.100	1.120 $\pm$ 0.095	1.603 $\pm$ 0.045
Global compact. $\uparrow$	1.192 $\pm$ 0.124	1.313 $\pm$ 0.465	2.427 $\pm$ 0.226
Per-pt. compact. $\uparrow$	3.273 $\pm$ 0.433	2.178 $\pm$ 1.148	6.396 $\pm$ 0.590
AUROC $\uparrow$	0.989 $\pm$ 0.003	0.983 $\pm$ 0.009	0.980 $\pm$ 0.012

than collapsing them into a shared class prototype. The AF cohesion remains balanced (0.846, vs. 0.921 SupCon, 0.955 BCE), indicating that the SR gains do not come at the cost of disordering the AF cluster. The objective preserves both classes’ per subject structure, in contrast to SupCon’s lopsided trade-off.

The BCE paradox. BCE produces the most clearly separated class means in the embedding space (centroid distance 1.603, cosine similarity -0.717) and the highest global compactness ratio (2.427). However, it gives the worst downstream AUROC on unseen patients (Section 3.2). The reason becomes clear once we look inside each patient. Those well separated class means coexist with disorganized per-patient SR placement (0.772 cohesion). A linear probe trained on seen patients has no consistent SR direction to extrapolate from when it meets a new one. This is consistent with the hypothesis that what governs transfer to new subjects is per-subject consistency, not how cleanly the classes are separated overall.. BCE achieves the highest *per-patient* compactness ratio (6.396). However, it is only because its denominator, the within-patient spread of each class shrinks toward zero. The positions of those tight clusters then drift unpredictably from one patient to the next, which is what the linear probe actually has to follow.

SupCon over-compacts AF, under-aligns SR. SupCon achieves the tightest AF cohesion (0.921) by pulling all AF segments together regardless of subject, at the direct cost of SR cohesion (0.800 vs. 0.850). This is the predicted failure mode. Standard supervised contrastive learning trades per-patient SR structure for global AF compactness, precisely the tradeoff that hurts patients with atypical SR baselines.

3.2. Downstream PAF Detection

The geometric ordering is preserved on the downstream task (Figure 2). AUROC orders the three losses identically to per-patient SR cohesion (Section 3.1). The most striking effect is on stability. Seed to seed AUROC variance is reduced by $2.6\times$ over SupCon and $3.4\times$ over BCE. A geometrically consistent embedding space is less sensitive

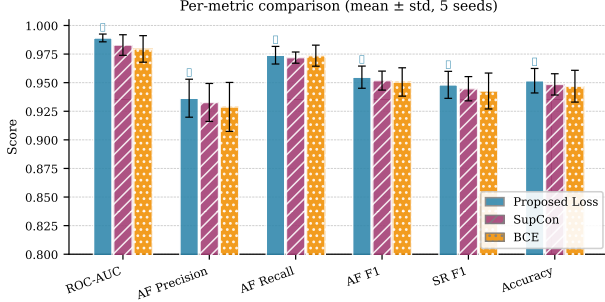


Figure 2. Per-metric linear-probe comparison of the proposed patient-aware objective, SupCon, and BCE on the held-out patient-independent test split (5 seeds). Bars show the seed mean and error bars show $\pm$ one standard deviation.

to weight initialization, which is a desirable property for any deployment that must reproduce model behaviour. AUROC improvements of $+0.006$ over SupCon and $+0.009$ over BCE each exceed one standard deviation of the corresponding baseline. Per-class precision/recall and a comparison to prior RRI-based AF detectors are reported in Appendix E (Tables 3 and 5). The proposed representations exceed Andersen et al.’s patient-holdout sensitivity (Andersen et al., 2019) by over 11 percentage points despite using only a frozen encoder and a logistic-regression probe.

4. Discussion

Per-subject consistency, not global separability, governs transfer. The BCE paradox is the clearest demonstration of our central claim. Class separability metrics tell us how far the class means have been pushed apart, but a linear probe trained on seen patients can only generalize to new ones if the embedding space provides a consistent direction along which to extrapolate. When the per-patient SR structure is disorganized, that direction is no longer well-defined even if the class means themselves are far apart. The proposed objective targets this consistency directly by forming positives only from same-patient, same-class segments, and its $+0.050$ SR-cohesion advantage over SupCon translates into a $2.6\times$ reduction in AUROC variance across seeds. Geometrically, the loss maintains a per subject balance between alignment and uniformity. Each patient’s class-conditional cluster is tightly aligned while clusters from different patients remain spread out on the hypersphere. This avoids both SupCon’s collapse into a single shared class prototype and BCE’s cross-patient inconsistency.

Generality and clinical relevance. The construction does not depend on the encoder and only requires subject identifiers and class labels in each batch. The mechanism applies wherever (a) class labels are available at training time and (b) within-subject variation is informative. Whether the same gains transfer to other physiological signals organized

by subject (EEG, EMG, PPG) remains an open empirical question and a direction for future work. On the clinical side, the consistent AF recall (0.974 ± 0.008) across seeds matters. A missed AF episode may delay anticoagulation and elevate stroke risk, so seed-stable sensitivity is a deployment advantage. The gain is in the representation, not the classifier. This downstream performance is achieved with a frozen encoder and a logistic regression probe and exceeds Andersen et al.’s patient-holdout sensitivity (Andersen et al., 2019) by over 11 percentage points (Appendix E, Table 5).

Limitations. Our analysis is on a single dataset (IRIDIA-AF) with a frozen-probe protocol. Prospectively, multi-centre validation and end-to-end fine-tuning remain open. Three concrete next steps follow from the geometric findings.

- A per-subject decomposition of alignment and uniformity to verify the mechanism formally.
- Few shot adaptation that uses each patient’s preserved SR structure as a personal prior.
- Applying the same class and subject positive construction to other physiological signals where between subject variability is the main barrier to generalization.

5. Conclusion

We proposed a patient-aware contrastive objective for physiological signals that are organized by subject. Positives are restricted to same-patient, same-class pairs, preserving each subject’s own structure while still pushing the two classes apart. On IRIDIA-AF, the construction attains the most consistent per-patient SR cohesion among the losses we compared, uncovers a BCE paradox. Overall class separability is a misleading indicator of how well a representation transfers to unseen subjects. Validated downstream, the construction reaches AUROC 0.989 ± 0.003 for PAF detection on unseen patients with $2.6\times$ lower seed variance than supervised contrastive baselines. The objective does not depend on the encoder and only requires subject IDs at training time. This suggests same-class, same-subject positive construction as a broadly useful primitive for representation learning on signals that are organized by subject.

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A. Preprocessing Pipeline

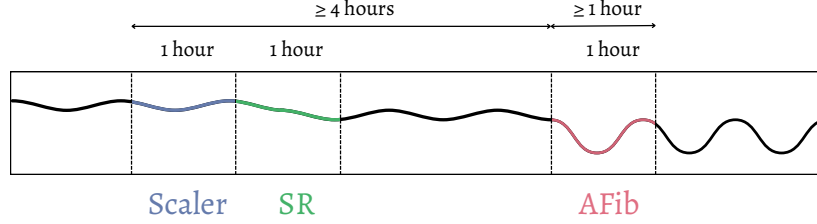


Figure 3. Episode selection and preprocessing pipeline. SR normalization windows (hour 0–1) are disjoint from classification windows (hours 1–2), ensuring the RobustScaler has no access to labeled data. The ≥ 4 hr SR inclusion criterion further ensures that all SR classification windows begin at least 2 hr before AF onset, reducing the risk of including pre-episode transitional rhythms.

Per-patient normalization uses,

$$\tilde{RR}_i = \frac{RR_i - \text{median}(\mathcal{R}_{\text{scale}}^{(p)})}{\text{IQR}(\mathcal{R}_{\text{scale}}^{(p)})} \quad (4)$$

where $\mathcal{R}_{\text{scale}}^{(p)}$ is the patient specific fit window (the first SR hour). Beats with $RR < 200$ ms or > 2000 ms are excluded as physiologically implausible. The hyperparameter optimization is described in Appendix B.

B. Encoder Architecture and Training Details

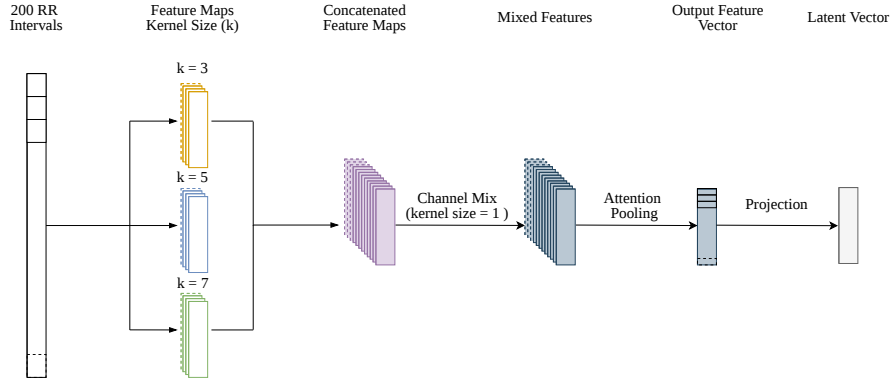


Figure 4. Encoder architecture: multi-branch 1D-CNN backbone with temporal attention pooling and a two-layer MLP projection head.

Architecture. Three parallel 1D-CNN branches with kernel sizes $k \in \{3, 5, 7\}$ capture beat-to-beat fluctuations, medium-range oscillations, and broader trend dynamics, motivated by Inception style multi-scale processing (Szegedy et al., 2015). Each branch applies three strided convolution blocks (stride 2) with channel depths $\{16, 32, 64\}$, Group Normalization (Wu & He, 2018) (8 groups), and ReLU. Branch outputs are concatenated and fused via a 1×1 convolution, then pooled by a softmax temporal attention module (Shashikumar et al., 2018),

$$\alpha_t = \frac{\exp(\mathbf{w}^\top \mathbf{h}_t)}{\sum_{t'} \exp(\mathbf{w}^\top \mathbf{h}_{t'})}, \quad \mathbf{z} = \sum_t \alpha_t \mathbf{h}_t. \quad (5)$$

A two-layer MLP projects $\mathbf{z} \in \mathbb{R}^{128}$ to $\mathbf{e} \in \mathbb{R}^{128}$; both are ℓ_2 -normalized to the unit hypersphere (Wang & Isola, 2020). The contrastive loss operates on $\hat{\mathbf{e}}$; the linear probe on $\hat{\mathbf{z}}$.

Training. PyTorch on a single NVIDIA RTX 2080. AdamW with learning rate 6.8×10^{-3} and weight decay 8.8×10^{-4} ; cosine annealing to 10^{-6} ; dropout 0.12 on the projection head; learnable temperature τ initialized at 0.05. Patient-aware

sampling uses $P = 8$ patients and $n = 16$ windows per class per-patient ($B = 128$). Up to 100 epochs with early stopping on validation AUROC (patience 10).

Hyperparameter selection. 32 hyperparameters (architecture, optimizer, data pipeline including W and S) are jointly tuned by Optuna’s TPE sampler (Akiba et al., 2019) on the validation split, with the test split held out until final evaluation. For all three loss conditions (Proposed, SupCon, BCE), only the temperature τ and learning rate are varied to find the best performance per loss. The encoder architecture, sampler, and probe are held fixed across all conditions, giving no tuning advantage to the proposed loss.

C. Embedding-Geometry Metric Definitions

The encoder produces L2-normalized embeddings, so every embedding $z_i \in \mathbb{S}^{d-1}$ satisfies $\|z_i\| = 1$. Let $y_i \in \{\text{SR}, \text{AF}\}$ denote the class label and $p_i \in P$ the patient identifier. Define the index sets

$$S^c = \{i : y_i = c\}, \quad S_p^c = \{i : y_i = c, p_i = p\},$$

the (unnormalized) class centroids

$$\mu^c = \frac{1}{|S^c|} \sum_{i \in S^c} z_i, \quad \mu_p^c = \frac{1}{|S_p^c|} \sum_{i \in S_p^c} z_i,$$

their L2-normalized versions $\hat{\mu}^c = \mu^c / \|\mu^c\|$, $\hat{\mu}_p^c = \mu_p^c / \|\mu_p^c\|$, and the mean class spreads

$$\bar{d}^c = \frac{1}{|S^c|} \sum_{i \in S^c} \|z_i - \mu^c\|_2, \quad \bar{d}_p^c = \frac{1}{|S_p^c|} \sum_{i \in S_p^c} \|z_i - \mu_p^c\|_2.$$

Per-patient class cohesion (primary metric). For class $c \in \{\text{SR}, \text{AF}\}$,

$$\text{Coh}^c = \frac{1}{|P|} \sum_{p \in P} \frac{1}{|S_p^c|} \sum_{i \in S_p^c} z_i^\top \hat{\mu}_p^c. \quad (6)$$

Because $\|z_i\| = \|\hat{\mu}_p^c\| = 1$, each summand is the cosine similarity between an embedding and its same-patient same-class centroid; values lie in $[-1, 1]$ and approach 1 as the per-patient class cluster tightens.

Global class separability.

$$\text{CentDist} = \|\mu^{\text{SR}} - \mu^{\text{AF}}\|_2, \quad (7)$$

$$\text{CentSim} = \hat{\mu}^{\text{SR}^\top} \hat{\mu}^{\text{AF}}, \quad (8)$$

$$C_{\text{glob}} = \frac{\|\mu^{\text{SR}} - \mu^{\text{AF}}\|_2}{\bar{d}^{\text{SR}} + \bar{d}^{\text{AF}}}. \quad (9)$$

We refer to C_{glob} as a *compactness ratio*. It is the ratio of between-class centroid distance to the sum of within-class mean spreads.

Per-patient compactness.

$$C_{\text{pp}} = \frac{1}{|P|} \sum_{p \in P} \frac{\|\mu_p^{\text{SR}} - \mu_p^{\text{AF}}\|_2}{\bar{d}_p^{\text{SR}} + \bar{d}_p^{\text{AF}}}. \quad (10)$$

C_{pp} measures the same separation-to-spread trade-off as C_{glob} , but evaluated within each patient and then averaged.

Note: when within-patient class spread is very small ($\bar{d}_p^{\text{SR}} + \bar{d}_p^{\text{AF}} \rightarrow 0$), the ratio becomes large regardless of whether the resulting clusters are positioned consistently across patients. In such cases C_{pp} should be interpreted alongside the raw spreads $\bar{d}_p^c$ and the per-patient cohesion scores rather than in isolation.

Table 2. Embedding-space metrics on the test set (mean $\pm$ std, 5 seeds). Per patient SR cohesion is the primary metric. Bold = best per row. $\uparrow$ higher is better; $\downarrow$ lower is better.

Metric	Proposed	SupCon	BCE
SR cohesion $\uparrow$	0.850 ± 0.044	0.800 ± 0.029	0.772 ± 0.049
AF cohesion $\uparrow$	0.846 ± 0.048	0.921 ± 0.035	0.955 ± 0.019
Centroid similarity $\downarrow$	$+0.238 \pm 0.200$	$+0.155 \pm 0.123$	-0.717 ± 0.029
Global centroid dist. $\uparrow$	1.034 ± 0.100	1.120 ± 0.095	1.603 ± 0.045
Global compactness ratio $\uparrow$	1.192 ± 0.124	1.313 ± 0.465	2.427 ± 0.226
Per-patient compactness ratio $\uparrow$	3.273 ± 0.433	2.178 ± 1.148	6.396 ± 0.590
AUROC $\uparrow$	0.989 ± 0.003	0.983 ± 0.009	0.980 ± 0.012

Table 3. Per-class linear-probe metrics for the proposed representations (4160 AF and 3799 SR segments from unseen patients; mean $\pm$ std across 5 seeds). The headline AUROC of **0.989 ± 0.003** is reported in the main text.

Class	Prec.	Recall	F1
SR (0)	0.970 ± 0.009	0.927 ± 0.020	0.948 ± 0.012
AF (1)	0.936 ± 0.017	0.974 ± 0.008	0.955 ± 0.010
Accuracy		0.952 ± 0.011	
AUROC		0.989 ± 0.003	

D. Embedding-Geometry Metrics (Full Table)

E. Downstream PAF Detection (Per-Class Metrics and Loss Comparison)

F. Comparison with Prior Work

Table 4. Linear-probe comparison across loss functions on the fixed patient-independent test split (5 seeds; bold = best mean per metric).

Loss	AUROC	AF Prec.	AF Recall	AF F1	SR F1	Acc.
Proposed	0.989±0.003	0.936±0.017	0.974±0.008	0.955±0.010	0.948±0.012	0.952±0.011
SupCon	0.983±0.009	0.933±0.017	0.972±0.005	0.952±0.008	0.945±0.011	0.949±0.009
BCE	0.980±0.012	0.929±0.021	0.974±0.009	0.951±0.012	0.943±0.016	0.947±0.014

Table 5. Performance comparison with related work on RR-interval AF detection. Direct numerical comparison is limited by dataset and evaluation differences. [†] Cross-validation with within-patient data mixing inflates reported metrics. [‡] MIT-BIH AF contains no episodes satisfying our quality criteria (AF $\geq$ 1h, preceding SR $\geq$ 4h).

Study	Method	Dataset	Validation	Sens.	Spec.	Acc.
(Udawat & Singh, 2022)	HRV+ML	MIT-BIH AF	CV [†]	95.16%	92.46%	94.43%
(Andersen et al., 2019)	CNN+RNN	3 dbs	5-fold CV [†]	98.98%	96.95%	—
(Andersen et al., 2019)	CNN+RNN	Unseen	Pt. holdout	86.04%	98.96%	—
(Faust et al., 2018)	LSTM	MIT-BIH [‡]	10-fold CV [†]	—	—	98.51%
Ours	CNN+MLP	IRIDIA-AF	Pt. holdout	97.40±0.80%	92.72±2.01%	95.17±1.07%